EHR Media Intelligence Platform

Design Write-up

Devansh Thakkar · AI Full-Stack Internship Assessment · github.com/DevanshThakkar19/ehr-media-intelligence

Problem framing

Clinical media in Epic-like environments—labs, imaging notes, discharge summaries—rarely sits in clean structured fields. The failure mode is not "missing AI"; it is expensive human scanning across inconsistent exports. This system treats that as a *data pipeline with a thin retrieval product*: ingest synthetic heterogeneous media, normalize to a testable intermediate representation (IR), emit validated HL7 FHIR R4 Bundles, generate orientation-only summaries, and expose sub-second semantic search in a clinician UI.

Architecture

Ingestion → IR → FHIR → Summary → Index → UI. Dirty CSV/JSON/plaintext never maps straight into FHIR; cleaning lives in Pydantic models with an audit trail. FHIR R4 emits `Patient`, `DocumentReference`, and `DiagnosticReport` as `Bundle.type=collection` (read/search, not write-back). Summaries cache on `MRN + bundle hash`. Search uses `all-MiniLM-L6-v2` + ChromaDB with resource-type and date filters. UI is Tailwind + vanilla JS over FastAPI—minimal setup, explicit API contract.

Tradeoffs and why

- **IR before FHIR.** Direct messy-to-FHIR mapping hides quality defects under coding systems. One hop buys unit-testable MRN/DOB/gender/date normalization, dedupe, and conservative identity resolution (fill missing MRN on name+DOB; never merge conflicting MRNs).
- **Collection Bundles, not transactions.** Assessment scope is retrieval, not Epic write-back—collection is the correct carrier for a patient's media set.
- **SQLite + Chroma.** `pip install` + `uvicorn` for reviewers; ∼50 records do not justify a service mesh.
- **Mock summarizer by default.** Live OpenAI/Anthropic are config flips; offline review must not depend on vendor keys.
- **Narrative in `description`/`conclusion`.** Searchable text over opaque Attachments at this size; production would still retain Attachments as legal source of truth.
- **Vanilla JS over React.** Matches the brief. A production console would still use a design-system SPA with authZ and audit.

FHIR / clinical research

FHIR R4 reference integrity (`subject`, `encounter`); `collection` vs. `transaction`; LOINC document-type codes for discharge, imaging, lab, and progress notes; administrative gender vs. clinical sex. Summaries are labeled **orientation only**—not a decision aid—and must not invent diagnoses absent from source media.

Validating AI summary quality

1. **Contract tests:** structured sections, ≤200 words, mandatory disclaimer, cache hit on identical `MRN + bundle hash`.
2. **Faithfulness spot-checks** (CHF/BNP, nodule, appendicitis US, CAP, allergy): claims must appear in source text; temperature 0.2 + anti-fabrication prompt with a live key.
3. **Retrieval sanity:** "elevated BNP failure" ranks true media (e.g., BMP & BNP) above twin FHIR projections and unrelated summaries.

What I would ship next

US Core + real `Encounter` resources; PDF/OCR ingest; automated faithfulness (claim ↔ source-span); hybrid BM25+vector for MRN/accession exact match; AuthN/Z and immutable audit for a hospital VPC—the gap between a strong demo and something behind a BAA.